

FATE Report 2026

Technical Flow Documentation

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1 High-Level Operational Overview

The following diagram summarizes the production workflow from environment configuration to final report generation. It represents the logical sequence executed during a standard monthly production run.

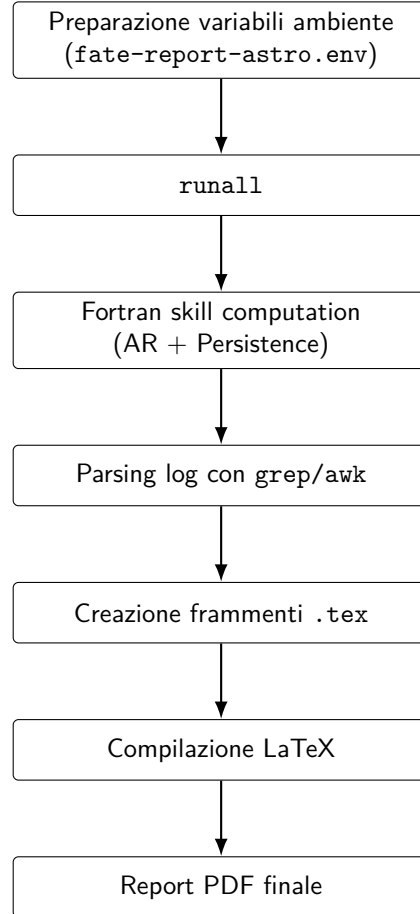


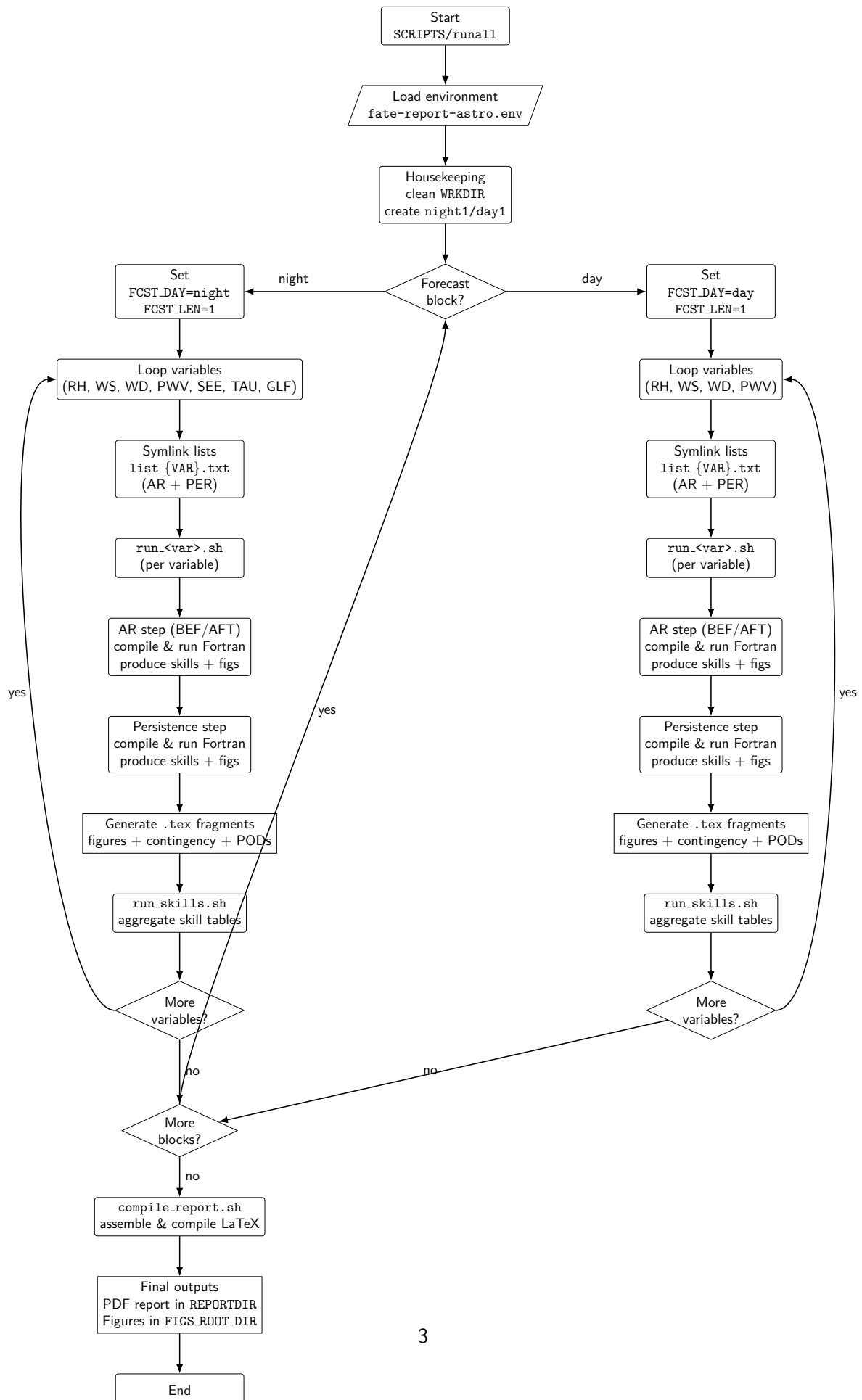
Figure 1: High-level operational workflow of the FATE reporting pipeline.

This diagram abstracts from implementation details and highlights the sequential nature of the processing chain. Each block corresponds to a well-defined operational stage with specific input/output artifacts.

2 Overview

- The technical execution flow of `SCRIPTS/runall`
- The interaction between shell scripts and Fortran programs
- The parameters configurable in `fate-report-astro.env`
- The impact of each parameter on the operational pipeline

3 High-Level Execution Flow



1. Entry Point

- `runall` is executed
- It automatically loads: `$HOME/NEW2026/SCRIPTS/fate-report-astro.env`

2. Environment Loading

The environment file `$HOME/NEW2026/SCRIPTS/fate-report-astro.env` defines:

- Data directories
- Working directory
- Forecast configuration
- Compilation paths
- Date ranges
- Report template paths

3. Housekeeping

- Clean `WRKDIR`
- Remove previous `.tex`, `.log`, `.aux`, `.png`, skills files
- Create subdirectories: `night1`, `day1`

4. Variable Loop

For each forecast block (e.g. `NIGHT1`, `DAY1`):

- Set `fcst_target` and `fcst_length`
- Loop over variables: `RH`, `WS`, `WD`, `PWV`, `SEE`, `TAU`, `GLF`
- Execute `run_<var>.sh`
- Execute `run_skills.sh`

5. Inside `run_<var>.sh`

1. Source environment and `functions.sh`
2. Retrieve variable attributes via `get_var_attr()`
3. Define working directory: `WRKDIR/{FCST_DAY}{FCST_LEN}`
4. Compile Fortran program via `gfortran`
5. Run executable with heredoc input parameters
6. Produce:
 - `skills_BEFAFT_<prefix>`
 - `skills_PER_<prefix>`
 - PDF figures

7. Parse skills files using `grep/awk`
8. Generate LaTeX fragments and tables:
 - `contingency_tableBEF*.tex`
 - `contingency_tableAFT*.tex`
 - `tablePODs*.tex`
 - `figures_<prefix>.tex`

6. `compile_report.sh`

- Concatenate generated `.tex` fragments
- Insert static sections (Introduction, References, Logs)
- Compile LaTeX into final PDF
- Output to `REPORTDIR`

5 `fate-report-astro.env` Configuration

5.1 Forecast Type

`FCST_TYPE`

- Allowed: `10min` or `mean`
- Determines `DATA_ROOT_DIR` and `DATA_PERS_DIR`

5.2 Scientific Parameters

<code>GG</code>	Scenario identifier passed to Fortran
<code>HH</code>	Time resolution identifier (e.g. <code>1H</code>)
<code>STARTMINUTE</code>	Start minute for statistics window
<code>ENDMINUTE</code>	End minute for statistics window
<code>LIMIT</code>	Threshold parameter passed to Fortran
<code>TAIL</code>	File extension (typically <code>.dat</code>)

5.3 Directories

<code>PROG_ROOT_DIR</code>	Location of AR Fortran programs
<code>PERS_ROOT_DIR</code>	Location of Persistence Fortran programs
<code>LIBPERSO_DIR</code>	Custom library path for compilation
<code>NUMREC_DIR</code>	Numerical Recipes library path
<code>WRKDIR</code>	Working directory for intermediate files
<code>FIGS_ROOT_DIR</code>	Output directory for figures
<code>REPORTDIR</code>	Final PDF output directory

5.4 Date Configuration

Defines report period and affects captions and filenames:

- STARTOFSERVICE_YYYY/MM/DD
- ENDOFSERVICE_YYYY/MM/DD
- Automatically derived date strings

6 Critical Operational Parameters

Before running `runall`, verify:

1. FCST_TYPE consistency with data directories
2. PROG_ROOT_DIR contains required .f90 files
3. Libraries compile correctly
4. LIST_* files exist and are coherent with the dates you want¹
5. WRKDIR and REPORTDIR have write permissions

7 Operational Summary

1. Configure `fate-report-astro.env`
2. Ensure data and list files are correct
3. Execute: `./runall`
4. Review output in WRKDIR and REPORTDIR

¹This step is crucial

8 Production-Ready Operational Checklist

This checklist is intended for operational use before executing `SCRIPTS/runall` in a production environment.

8.1 1. Environment Configuration Validation

Verify `FCST_TYPE` is correct (10min or mean).

Confirm `STARTMINUTE` and `ENDMINUTE` match operational window.

Confirm `GG` and `HH` are consistent with data resolution.

Validate service date range:

`STARTOFSERVICE_YYYY/MM/DD`

`ENDOFSERVICE_YYYY/MM/DD`

8.2 2. Directory Integrity

`PROG_ROOT_DIR` exists and contains required `.f90` files.

`PERS_ROOT_DIR` exists and contains persistence programs.

`LIBPERSO_DIR` and `NUMREC_DIR` exist and are accessible.

`DATA_ROOT_DIR` exists for selected `FCST_TYPE`.

`DATA_PERS_DIR` exists for selected `FCST_TYPE`.

`WRKDIR` exists and is writable.

`FIGS_ROOT_DIR` exists and is writable.

`REPORTDIR` exists and is writable.

8.3 3. List File Consistency

AR list files exist in `$PROG_ROOT_DIR`.

PERSISTENCE list files exist in `$PERS_ROOT_DIR`.

Symlinked `list_<VAR>.txt` matches current reporting month.

Number of nights (`wc -l`) consistent with expected sample size.

8.4 4. Compilation Validation

`gfortran` available in `PATH`.

Test compile of one Fortran program successful.

Linked libraries resolve correctly (no missing symbols).

8.5 5. Runtime Dry-Run Test

Execute one variable only (e.g. RH night1).

Verify generation of:

- skills_BEFAFT file
- skills_PER file
- PDF figures
- LaTeX fragments

Check parsing of contingency tables (no empty values).

8.6 6. Full Pipeline Execution

Execute `./runall`

Monitor for compilation errors.

Confirm no missing skills files.

Confirm all expected `.tex` fragments created.

8.7 7. Report Validation

Final PDF generated in `REPORTDIR`.

Figures correctly embedded.

Contingency tables populated.

POD tables consistent with skills files.

Date strings correct in captions.

8.8 8. Post-Run Archiving

Archive generated list files.

Archive environment file used for this run.

Tag report version (e.g. `YYYYMM`).

8.9 Production Readiness Gate

The run can be considered production-valid only if:

- No compilation errors occurred.
- No missing skills files were detected.
- All tables contain non-empty numerical values.
- The final PDF passes visual validation.

9 Directory Structure (Filesystem Representation)

The operational directory structure is reported below using a filesystem-style representation. Each block includes a short description of its purpose.

Filesystem Tree

```
NEW2026/
|
|---- SCRIPTS/
|   |---- runall                      # Main orchestration script
|   |---- run_<var>.sh                # Variable-specific execution scripts
|   |---- run_skills.sh              # Skill aggregation
|   |---- compile_report.sh          # LaTeX compilation and report assembly
|   '---- fate-report-astro.env      # Environment configuration
|
|---- PROG/
|   |---- TEST_AR/
|   |   |---- AR_10min/ORIGINAL/      # Fortran AR codes (.f90)
|   |   '---- PERSIST/ORIGINAL/      # Fortran persistence codes (.f90)
|   |---- NUMREC/                    # Numerical Recipes library
|   '---- LIBPERSO/                  # Custom compiled libraries
|
|---- DATA/
|   '---- TREATED/
|       '---- FATE_10min_AR/
|           |---- AR/
|               |---- night1/<VAR>_TREATED/*.dat
|               '---- day1/<VAR>_TREATED/*.dat
|           '---- PERSIST/
|               |---- night1/<VAR>_TREATED/*.dat
|               '---- day1/<VAR>_TREATED/*.dat
|
|---- LIST_DIR_AR/
|   |---- list_<VAR>.txt
|   '---- list_<VAR>_LastMonth.txt
|
|---- LIST_DIR_PER/
|   |---- list_<VAR>.txt
|   '---- list_<VAR>_LastMonth.txt
|
|---- WRKDIR/
|   |---- night1/
|       |---- skills_BEFAFT_<var>*
|       |---- skills_PER_<var>*
|       |---- contingency_table*.tex
|       |---- tablePODs*.tex
|       |---- table_skills_*.tex
|       '---- figures_<var>.tex
|   '---- day1/
|       |---- skills_BEFAFT_<var>*
```

```

|         |---- skills_PER_<var>*
|         |---- contingency_table*.tex
|         |---- tablePODs*.tex
|         |---- table_skills_*.tex
|         '---- figures_<var>.tex
|
|---- FIGS/
|   |---- night1/
|   |   |---- <var>_..._BEF.pdf
|   |   |---- <var>_..._AFT.pdf
|   |   '---- <var>_..._per.pdf
|   |
|   '---- day1/
|   |   |---- <var>_..._BEF.pdf
|   |   |---- <var>_..._AFT.pdf
|   |   '---- <var>_..._per.pdf
|
|---- REPORT/
|   |---- FATE-REPORT_<YYYYMM>.pdf
|   '---- Annex_<YYYYMM>.pdf

```

Interpretation

- **SCRIPTS/** contains orchestration logic and configuration.
- **PROG/** contains Fortran scientific computation codes and libraries.
- **DATA/** contains treated AR and Persistence datasets.
- **LIST_DIR_AR / LIST_DIR_PER** control which nights/files are processed.
- **WRKDIR/** contains intermediate products (skills files and LaTeX fragments).
- **FIGS/** contains generated PDF figures for each variable and block.
- **REPORT/** contains final production-ready PDF reports.